ASTANA IT UNIVERSITY
School of Software Engineering

FIRST SUBMISSION REPORT
(interim diploma project report)

Development of Web Application for Enhancing Digital Competencies of School Educators

Student Ayan Abilbek, SE-2401

Supervisor Aliya Katyetova

Astana 2026

ABSTRACT

Report: 59 pages, 5 main-text figures, 4 appendix figures, 22 main-text tables, 1 appendix table, 27 references and 3 appendices.

DIGITAL COMPETENCE, SCHOOL EDUCATORS, WEB APPLICATION, DIAGNOSTIC ASSESSMENT, PRACTICAL TASKS, USABILITY

This report presents the analysis and design of a web application for school educators. Its purpose is to connect an initial diagnostic activity with three focused learning modules, practical submissions, feedback and progress tracking. The work uses a critical literature review, a comparison of existing systems and a comparison of technical alternatives to define an initial specification. The proposed design contains teacher and administrator roles, versioned learning content, rule-based module recommendations and rubric-based review. A local interactive prototype makes the main learning sequence available for inspection. A testing and evaluation plan defines functional checks, data-protection checks, usability measures, parallel practical tasks and a static resource baseline. The project repository and tracker record the current work and the remaining dependencies. The report contains prepared design material; it does not report an implemented production application or measured learning gains. Discussions with educators, substantive supervisor feedback, recipient-access checks and a final submission receipt remain outstanding. These gaps must be resolved before the package can be treated as a complete first submission.

CONTENTS

ABSTRACT 2

ABBREVIATIONS 5

INTRODUCTION 6

Source of the submission rules 7

1 Project proposal 8

1.1 Project and responsibility 8

1.2 Problem and intended users 8

1.3 Aim and research question 8

1.4 Scope and practical contribution 9

1.5 Three modules and competence mapping 10

2 Evidence of relevance 11

2.1 Discussion protocol ready for use 11

2.2 How evidence will change the design 12

3 Literature review 14

3.1 Selection and limits 14

3.2 Competence is broader than operating software 14

3.3 Self-assessment and practical performance need separate records 15

3.4 Training must lead to a task that can be examined 15

3.5 Current frameworks and accessibility 16

3.6 Structured analysis of the core sources 16

3.7 Resulting design gap 17

4 Comparison of existing systems 18

5 Technology comparison and decisions 20

5.1 Decision A Server platform 20

5.2 Decision B Data storage 20

5.3 Decision C Component interaction 21

5.4 Recommendation approach 22

6 Requirements specification 23
6.1 Non-functional requirements 24
6.2 Scoring and module selection 25
6.3 Main use case and alternatives 26
7 Architecture and diagrams 27
7.1 Responsibilities and trust boundaries 29
7.2 Initial data model 30
7.3 Request and failure handling 31
8 UX prototype 32
8.1 Interaction decisions to validate 34
9 Verification and evaluation plan 35
9.1 Questions and baseline 35
9.2 Measurement conditions 36
9.3 Practical rubric and bias control 37
9.4 Release gates 38
10 Project management and risks 39
10.1 Work breakdown and task traceability 40
10.2 Risks and responses 40
10.3 Repository and tracker acceptance audit 42
11 Supervisor feedback 43
CONCLUSION 44
REFERENCES 45
APPENDIX A 49
APPENDIX B 54
APPENDIX C 56

ABBREVIATIONS

API - application programming interface

CSV - comma-separated values

DigCompEdu - European Framework for the Digital Competence of Educators

FR - functional requirement

HTTPS - Hypertext Transfer Protocol Secure

NFR - non-functional requirement

PR - pull request

REST - representational state transfer

TC - test case

UI - user interface

UX - user experience

WCAG - Web Content Accessibility Guidelines

INTRODUCTION

This report presents the analysis and proposed design for an individual diploma project. It is a review draft revised on 23 October 2026 for the first submission due on 24 October 2026 at 18:00, Astana time. It covers all eleven required components. The design sections are developed, but the package is not yet a complete submission: discussions with users, substantive supervisor feedback, recipient access and the submission receipt remain outstanding.

The project title and initial scope follow the final Justification dated 15 October 2026. In an email dated 16 October, Aliya Katyetova confirmed that she had completed, signed and sent it to the School of Software Engineering. This existing document is preserved without revision. The locally available PDF has an empty visible signature line and no PDF signature field; the email is the available confirmation of signing and dispatch. This distinction does not call for rewriting the justification.

Evidence labels used in this report have a narrow meaning. Confirmed means supported by a source document, correspondence or recorded technical check. Proposed means a design or planning decision prepared for review. Missing means the required evidence has not been supplied or produced. No teacher interviews, learning gains, application test results or supervisor approvals are asserted.

Table 1 maps the required components to the available material and the remaining evidence.

Table 1 - Coverage of the eleven required components

Component	Current status	Evidence or remaining work
1 Proposal	Prepared for review	Based on the final Justification; local needs still need validation.
2 Relevance evidence	Missing primary evidence	User confirmed that discussion records do not exist. Protocol provided.
3 Literature review	Prepared	12 core sources, including 8 scientific publications; access limitations stated.
4 Systems comparison	Prepared	3 systems compared on 8 criteria using official documentation.
5 Technology comparison	Prepared	3 decisions with 3 alternatives each; choices are proposed.

Continuation of Table 1

Component	Current status	Evidence or remaining work
6 Requirements	Prepared for validation	Numbered requirements, priorities and acceptance criteria.
7 Architecture	Prepared	Context, use case and component diagrams with editable sources.
8 UX	Prepared	Local interactive prototype with demonstration data. No user validation.
9 Verification plan	Prepared	Functional tests, outcome measures, baseline and measurement conditions.
10 Project management	Partial	GitHub and YouTrack created; milestones and risks prepared. Recipient access and supervisor review remain pending.
11 Supervisor feedback	Missing	Signed justification is not a review of these new deliverables.

Source of the submission rules

The local requirements DOCX specifies the eleven components and repository workflow. The School email forwarded on 16 October additionally requires a repository link, a YouTrack or similar project link, a PDF of board/task screenshots and a separate consolidated report. The supervisor also explicitly confirmed that GitHub is acceptable. University Microsoft authentication was completed and pages 1-2 of the live form were checked. The form explicitly requires a prob-period-1 tag URL, task IDs and commit or pull-request links for individual contributions. The DOCX uses a different tag name, pre-defense-1; both should point to the same final commit. Pages 3-5 and upload limits have not yet been checked because the required final tag is not ready. The authoritative form is https://forms.cloud.microsoft/r/Yy6AXT5pe3.

1 Project proposal

1.1 Project and responsibility

Student: Ayan Abilbek, Software Engineering, group SE-2401. Supervisor: Aliya Katyetova. This is an individual project. I am responsible for requirements, architecture, implementation, tests, evaluation and the thesis. The supervisor reviews the academic scope, competence mapping and evaluation approach. Her participation in reviews is not treated as implementation work.

1.2 Problem and intended users

A teacher preparing a lesson may need to choose a reusable image, make a short online quiz and share the resulting resources safely. Learning how to operate a tool is only part of that work: the teacher also has to decide whether the resource fits the lesson and whether it can be used responsibly. The project addresses this sequence through a small web application that connects an initial assessment to focused learning and practical tasks.

The intended users are school educators who want to improve selected digital teaching skills. An administrator maintains learning content and reviews practical submissions. Pupils do not need accounts, and their personal information is not needed for the pilot. The final Justification records a personal connection to school teaching through help with lesson materials. That experience motivates the project, but it does not establish the needs of a wider teacher population. Those needs remain to be checked in 3-5 discussions.

1.3 Aim and research question

The aim is to design, implement and evaluate a web application that helps school educators identify a limited set of learning needs, complete relevant learning activities and monitor their practice. The research question is: how can a short assessment, explainable recommendations and classroom-related practice be combined into a usable learning workflow for school educators? A later pilot will examine usability and preliminary changes in practical task performance. It will not establish national competence levels or causal effectiveness from a small convenience sample.

The objectives, deliverables and completion evidence are set out in Table 2.

Table 2 - Project objectives and deliverables

Objective	Deliverable	Evidence of completion
O1 Understand the problem	Discussion synthesis and competence map	3-5 documented discussions or substantive client agreement; assumptions separated.
O2 Specify and design	Requirements, three diagrams and UX prototype	Traceable scenarios and reviewable acceptance criteria.
O3 Implement the core workflow	Teacher/admin accounts, assessment, 3 modules, tasks and progress	End-to-end demonstration and automated integration checks.
O4 Evaluate usability and learning signals	Pilot protocol, anonymised observations and task scoring	Actual paired records, denominators and limitations reported.
O5 Maintain reproducible work	Repository, tracker and release documentation	Tasks linked to changes; selected supervisor reviews; tagged submission.

1.4 Scope and practical contribution

The initial version contains exactly three learning modules: creating digital learning resources; designing online assessment; and managing educational resources responsibly. A teacher completes an initial assessment, sees a suggested starting module and the reason for that suggestion, studies a short lesson, submits a practical response and checks progress. The administrator publishes versioned content and reviews submissions using a simple rubric. A teacher may choose another module instead of following a recommendation.

The proposed contribution is this small, traceable workflow and its evaluation with school educators. The project does not claim to invent digital competence assessment or to outperform established training platforms. It excludes an AI tutor, automatic grading of open-ended teaching quality, a course marketplace, live classes, school information system integrations and formal professional certification. These exclusions keep the individual project manageable.

1.5 Three modules and competence mapping

Table 3 connects each proposed module to a classroom task and a competence reference.

Table 3 - Learning modules and competence mapping

Module	Proposed activity	DigCompEdu reference and evidence
M1 Digital resources	Prepare a one-page learning resource for a stated lesson objective. Explain the choice of source, adaptation and accessibility.	2.1 Selecting; 2.2 Creating/modifying; 5.1 Accessibility/inclusion. Evidence: resource excerpt and rationale.
M2 Online assessment	Design a five-question formative quiz and explain how feedback would change the next lesson. Use fictitious responses.	4.1 Assessment strategies; 4.2 Analysing evidence; 4.3 Feedback/planning. Evidence: quiz specification and feedback decision.
M3 Responsible resource management	Choose a suitable sharing setting, attribute a reusable resource and remove personal data from a fictional example.	2.3 Managing/protecting/sharing resources; 6.3 Digital content creation. Evidence: explanation of permission, attribution and privacy choices.

The mapping uses DigCompEdu as a design reference [1]. It covers selected competences only; it is not an implementation of all 22 competences or a validated A1-C2 classification. The precise tasks and language must be checked with educators and the supervisor before recruitment and assessment.

2 Evidence of relevance

Primary user evidence is not yet available. The student confirmed that there are no discussion records. No participant counts, quotations, interview themes or local survey results are reported here. Literature supports the general problem, while the final Justification supplies the intended scope; neither replaces discussions with the intended users.

Table 4 distinguishes the available context from questions that still require user evidence.

Table 4 - Evidence and unresolved assumptions

Item	What is known	What remains unverified
Project context	The final Justification identifies school educators and three learning areas.	Whether these three areas are priorities for the teachers who will use the prototype.
Personal observation	The Justification records experience helping with lesson materials.	Frequency, severity and distribution of difficulties across educators.
General research	Reviewed studies discuss digital competence, training and assessment.	Transferability to the selected school context in Kazakhstan.
Proposed learning format	Short modules and practical tasks are planned.	Preferred language, available time, device access and acceptable feedback format.

2.1 Discussion protocol ready for use

Recruit 3-5 adult school educators with some variation in subject, teaching experience and confidence with technology. A small convenience sample is suitable for discovering concrete requirements, not estimating prevalence. If a family member participates, record that relationship and recruit other educators too. Each discussion should last around 20 minutes. Participation is voluntary; the educator may skip questions. Record audio only with explicit permission. Use participant codes in shared materials and keep identities outside the repository.

The discussion protocol is organised in Table 5.

Table 5 - Protocol for discussions with educators

Step	Question or activity	What to record
1 Context	What subject and age group do you teach? Which devices and tools do you normally use?	Role, device and tool context; no unnecessary personal details.
2 Recent task	Tell me about the last lesson resource or online quiz you prepared.	Concrete sequence, time-consuming steps and actual obstacles.
3 Current workaround	What did you do when you got stuck? Which help was useful?	Existing alternatives, workarounds and dissatisfaction.
4 Priorities	Which of the three proposed learning areas would be useful, and which would not? Why?	Supporting and contradicting evidence; participant wording.
5 Prototype walkthrough	Show how you would find a module, submit a task and view feedback.	Observed behaviour, hesitation and assistance; distinguish this from opinions.
6 Constraints	What language, session length and device would make this usable? What would you avoid sharing?	Requirements for language, time, access and privacy.
7 Confirmation	Summarise the main points and ask whether the interpretation is accurate.	Corrections and permission to use anonymised notes.

2.2 How evidence will change the design

For each observation, retain a source code, a short anonymised excerpt or faithful paraphrase, an interpretation and a linked requirement. Record contradictory accounts instead of forcing agreement. A requirement based on one educator should be marked as such. Counts such as "2 of 4 participants" may describe this sample, but must not be presented as percentages of all teachers. Update module priorities, terminology and accessibility assumptions after synthesis.

Table 6 records the main assumptions and the actions needed to check them.

Table 6 - Assumptions and validation actions

Assumption	Risk if wrong	Validation action
A1 Teachers will use short self-paced sessions.	The proposed lesson length may not fit their routine.	Ask about a recent learning episode and available time.

Continuation of Table 6

Assumption	Risk if wrong	Validation action
A2 Three selected topics are useful.	The application may solve a low-priority problem.	Have participants rank topics and name missing tasks.
A3 Teachers can access a browser and stable connection.	The workflow may exclude intended users.	Record device/network constraints before implementation.
A4 Russian content is an acceptable initial option.	Language could obstruct participation.	Confirm preferred language; keep interface strings externalised.
A5 An administrator can review practical tasks.	Feedback may be delayed or inconsistent.	Agree reviewer availability and calibrate the rubric.

3 Literature review

3.1 Selection and limits

This is a focused critical review for the first project submission, not a systematic review. Searches conducted on 22 October 2026 used combinations of "teacher digital competence", "DigCompEdu", "in-service teachers", "assessment" and "professional development". Publisher pages, original research reports and official framework documentation were checked. Eight scientific publications are retained, seven published in 2021-2025, together with four framework/standard sources. Falloon (2020) is retained for its conceptual treatment of professional and ethical competence. No fabricated screening counts or database access are claimed.

The review prioritises studies of school educators, instruments and training design. The higher-education review [2] is retained as a methodological comparison, with its different population made explicit. For [3] and [4], the available publisher abstract/highlights support only a bounded account; no inaccessible methods or effect sizes are attributed to them. Official technical and product documentation is listed separately and does not count toward the eight scientific publications.

3.2 Competence is broader than operating software

DigCompEdu provides a common structure for linking teaching activities to digital competence [1]. Its value for this project is traceability: a module can state the competence it addresses and the task through which a teacher practises it. UNESCO likewise treats its framework as something to adapt to a particular setting [5]. Neither framework establishes the needs of the teachers recruited for this project.

Falloon argues for a broader understanding that includes ethical and professional capabilities [6]. Heine, Krepf and König show that even "digital resources" is used inconsistently across research [7]. These points lead to a concrete design choice: M1 must ask why a resource is appropriate and reusable, rather than simply whether a teacher can upload a file. The rubric will state the intended lesson objective and the evidence required.

3.3 Self-assessment and practical performance need separate records

Lucas and colleagues validated a DigCompEdu-based self-assessment with 1,071 in-service teachers and examined personal and contextual factors [8]. Tzafilkou and colleagues developed a 20-item instrument using responses from 845 primary and secondary teachers in Greece [9]. These studies show how structured instruments can examine several dimensions. Their samples and validation procedures do not transfer automatically to a new, shorter questionnaire.

Basilotta-Gómez-Pablos and colleagues found a strong emphasis on self-assessment in the higher-education literature [2]. Claro and colleagues identify varying operational definitions and limited comparability in quantitative research on school teachers [4]. For this application, confidence responses will therefore be stored separately from scored scenario questions and practical task rubrics. A single combined "competence percentage" would conceal those differences. The prototype will not award a DigCompEdu proficiency level.

3.4 Training must lead to a task that can be examined

Cabero-Almenara and colleagues designed and validated the structure, content and tasks of a DigCompEdu-based t-MOOC for non-university teachers [3]. This is relevant to the organisation of modules, but validation of a training proposal does not prove that this application improves learning. Gallego Joya and colleagues review training programmes and note both reported improvements and gaps in alignment with current needs [10]. Together, these sources support testing a complete learning sequence and checking its fit with local educators.

The proposed sequence is: identify a learning need, explain the recommendation, provide a short lesson, complete a classroom-related task and inspect feedback. The learning evaluation will use parallel practical tasks before and after a module. Completion clicks are a usage measure; they are not evidence of acquired skill. The pilot must report assistance, missing responses and participant selection alongside any change in scores.

3.5 Current frameworks and accessibility

DigComp 3.0, published in 2025, adds more granular learning outcomes and integrates AI across the general citizen framework [11]. It is useful for checking whether resource-management examples ignore current digital practices. It does not replace the educator-specific scope approved in the Justification, and it does not require adding an AI product feature. WCAG 2.2 [12] supplies testable accessibility expectations for the interface, including keyboard use, focus visibility and readable controls. These become planned checks, not a claim of certified conformance.

3.6 Structured analysis of the core sources

The contribution and limits of each core source are compared in Table 7.

Table 7 - Critical analysis of the core literature

ID	Publication	Approach and contribution	Use and limitation
[1]	Redecker (2017), DigCompEdu	Framework for educators; six areas and 22 competences.	Use selected descriptors for the module map. A reference framework is not local needs evidence.
[5]	UNESCO (2018), ICT CFT v3	Framework intended for contextualised teacher development.	Validate local conditions and reviewer capacity. Does not validate this web application.
[6]	Falloon (2020), TDC framework	Conceptual article broadening technical skills toward professional and ethical competence.	Include rationale, responsible sharing and reflection. Conceptual basis, not an intervention effect estimate.
[8]	Lucas et al. (2021)	Empirical study; 1,071 in-service teachers; validated self-assessment and factor analysis.	Separate confidence from observed performance. Associations do not establish causal training effects.
[3]	Cabero-Almenara et al. (2021)	Design/validation of a task-oriented MOOC for non-university teachers; publisher abstract inspected.	Structure modules around tasks. Do not infer learning gains from design validation.
[2]	Basilotta-Gómez-Pablos et al. (2022)	Systematic review of 56 higher-education studies; self-assessment prominent.	Triangulate measures. University staff differ from the target school population.

Continuation of Table 7

ID	Publication	Approach and contribution	Use and limitation
[7]	Heine, Krepf and König (2023)	Review of digital-resource definitions in 23 articles.	Define the resource and pedagogical purpose explicitly. Terminology review is not a usability study.
[4]	Claro et al. (2024)	Review of quantitative school-teacher studies from 2016-2021; publisher abstract/highlights inspected.	State what each metric measures; avoid comparing unlike instruments or populations.
[9]	Tzafilkou et al. (2023)	Empirical instrument study; 845 Greek teachers, 20 items and six components.	Include professional and pedagogical scenarios. Translation and shortening require fresh validation.
[10]	Gallego Joya et al. (2025)	Review of training programmes in studies published in 2017-2023.	Link conceptual learning, practical tasks and attitudes; investigate local fit rather than assuming transfer.
[11]	Cosgrove and Cachia (2025), DigComp 3.0	Updated citizen competence framework; technology-neutral learning outcomes.	Check contemporary resource examples. Keep educator mapping grounded in [1].
[12]	W3C (2024), WCAG 2.2	Technical accessibility recommendation, not an empirical study.	Derive interface checks and document failures. Automated checks alone cannot establish full conformance.

3.7 Resulting design gap

The evidence supports a bounded engineering question: can a small application make the path from assessment to relevant practice clear and usable for a specific group of school educators? It does not support a claim that no comparable platform exists. The project will contribute an implemented workflow, transparent decision rules, traceable content and a cautious pilot evaluation. The local user discussions remain essential because the reviewed studies do not establish demand in the intended school setting.

4 Comparison of existing systems

Three relevant systems are compared using eight consistent criteria. The comparison uses official public documentation inspected on 22 October 2026; no paid accounts, hands-on usability scores or comparative experiments are claimed. "Not established" means that the inspected documentation did not establish the feature. It must not be read as proof that the feature is absent. Features and terms can change.

Table 8 applies the same eight criteria to three existing systems.

Table 8 - Comparison of existing systems

Criterion	SELFIE for TEACHERS [13]	Google for Education [14-15]	Microsoft Learn Educator Center [16-17]
Primary purpose	Reflection on digital teaching competence.	Training and certification in Google education tools.	Educator training and professional development.
Audience	Early years, primary and secondary teachers.	Educators and education administrators.	Educators; catalogue supports role filters.
Competence reference	Six teaching-related areas; reflection and feedback.	Product-oriented skills; selected standards-aligned offers.	Product and topic learning paths; full DigCompEdu map not established.
Initial assessment	Reflective questions and personalised report.	Certification exams documented; diagnostic-to-module matching not established.	Knowledge checks/modules documented; universal entry diagnosis not established.
Learning content	Improvement suggestions; a full practice course is not established on the inspected page.	Self-paced courses and practical preparation resources.	Interactive lessons, modules and learning paths.
Practical evidence	Self-reflection output; reviewed teaching artefact workflow not established.	Certification and task preparation; grading of this project’s local rubric not established.	Training activities; grading of this project’s local rubric not established.
Progress and output	Personal report, participation certificate/badge.	Educator certification and badge.	Training history, achievements and transcript sharing.

Continuation of Table 8

Criterion	SELFIE for TEACHERS [13]	Google for Education [14-15]	Microsoft Learn Educator Center [16-17]
Account and ecosystem	EU Login; optional sharing of personal report.	Google education ecosystem.	Microsoft Learn profile; educator/product catalogue.

SELFIE offers a strong reference for reflective feedback, while the Google and Microsoft services already provide extensive educator learning material. A claim that teachers lack any training platform would be inaccurate. The proposed application instead tests a narrow sequence across three locally chosen topics, with explicit recommendation reasons and practical evidence visible beside progress. Whether that sequence is more useful than a simple resource list is an evaluation question, not a conclusion from this table.

The pilot baseline will be a curated static resource page covering the same three topics and comparable content. It will be labelled as a project baseline, not as a reproduction of any commercial platform. The comparison isolates navigation and guidance more clearly than comparing different content libraries, although a small pilot will still have strong order and familiarity effects.

5 Technology comparison and decisions

The final Justification proposes Next.js, React, TypeScript and PostgreSQL. The following comparisons examine that choice against the project needs. They are design judgements, not performance benchmarks. Dependency versions and the deployment provider will be pinned when implementation begins. No cloud subscription or hosting cost is assumed.

5.1 Decision A Server platform

The server platform alternatives are compared in Table 9.

Table 9 - Server platform alternatives

Option	Fit for this project	Cost or limitation
Next.js with TypeScript [18]	UI and server route handlers in one project; shared language; sufficient for accounts, content and assessment.	Server/client boundaries and authorisation need care. Framework conventions do not provide secure access automatically.
Django [19]	Mature model-driven backend and built-in administration; suitable for content and relational data.	A separate rich React client would add a second language and integration work; server templates remain viable.
NestJS plus React [20]	Explicit modules, controllers and services; clear backend boundaries.	Separate frontend/backend setup and deployment add work for a small individual project.

Proposed decision: keep Next.js as a modular monolith. Business rules live in server-side services rather than UI components. This preserves a small deployment surface while leaving assessment, recommendations and progress independently testable. Reconsider a separate backend if the scope grows beyond the approved workflow.

5.2 Decision B Data storage

Table 10 compares the storage alternatives against the proposed data model.

Table 10 - Data storage alternatives

Option	Fit for this project	Cost or limitation
PostgreSQL [21]	Foreign keys and transactions suit attempts, responses, published module versions and progress updates.	Requires a running database, migrations and a tested backup process.
SQLite [22]	Simple local database; useful for small standalone tools and development experiments.	A shared hosted application with concurrent writes and remote storage needs a considered deployment model.
MongoDB [23]	Flexible documents can represent variable learning content.	Relational invariants and cross-entity reporting would need deliberate modelling; flexibility is not itself a requirement.

Proposed decision: use PostgreSQL for the shared application. Publish content versions and retain the version identifier with each assessment attempt. Save an attempt and its responses in one transaction so a failed request cannot leave a completed attempt without the corresponding data. Content documents may use a bounded JSON field, but ownership and version relationships remain explicit.

5.3 Decision C Component interaction

The interaction options and their costs are compared in Table 11.

Table 11 - Component interaction alternatives

Option	Fit for this project	Cost or limitation
HTTPS JSON endpoints	A small explicit API maps well to attempts, tasks, reviews and progress. Easy to exercise in integration tests.	Endpoint validation, consistent errors and server-side access checks must be implemented.
GraphQL [24]	Useful when several clients need varied data shapes from a shared schema.	Schema/resolver permissions and extra tooling bring limited benefit for one small web client.
Next.js Server Actions [25]	Convenient for form mutations inside the same application.	Tighter framework coupling; actions still need input validation and authorisation.

Proposed decision: use HTTPS JSON route handlers for the core workflows and direct server-side queries for read-only server rendering where appropriate. This keeps the acceptance tests independent of page structure. No microservice network, event broker or real-time channel is needed for the first version.

5.4 Recommendation approach

The approved scope calls for rule-based recommendations. Machine learning would require suitable training data and a defensible evaluation that are not currently available. A versioned rule sorts modules by diagnostic need and presents the reason to the teacher. The rules are simple enough to inspect, test at boundary values and change after feedback. They do not claim to predict an individual’s ability.

6 Requirements specification

The following requirements describe the intended application, not completed functionality. Must denotes essential scope, Should denotes useful scope that can be deferred. All requirements are proposed from the final Justification and the reviewed sources; none is labelled as validated by interviews. Teacher needs will be linked using evidence codes after the discussions. Administrator accounts belong to a designated content reviewer; supervisor and committee access to project materials does not automatically grant access to teachers' personal records.

Table 12 specifies the required functions, priorities and observable acceptance criteria.

Table 12 - Functional requirements and acceptance criteria

ID / priority	Requirement	Acceptance criteria
FR01 / Must	Account access and roles	A registered teacher can sign in/out. Unauthenticated requests receive 401 and a teacher cannot call administrator operations (403). Passwords are never stored in plaintext.
FR02 / Must	Initial assessment	A teacher completes 9 original scenario questions, 3 per module, plus separate confidence prompts. Missing mandatory answers prevent submission and receive a field-level error.
FR03 / Must	Scores and history	The server stores answers, score, timestamp and content/rule versions. Confidence is displayed separately. A repeated request with the same idempotency key creates one attempt.
FR04 / Must	Explainable recommendation	The lowest-scoring incomplete module is recommended; a tied score uses M1, M2, M3 order. The page names the topic and score behind the choice. With no assessment, show the catalogue without a personalised claim.
FR05 / Must	Three learning modules	Each published module contains an objective, competence reference, lesson, example, practice brief and completion rules. Unpublished versions are not visible to teachers.
FR06 / Must	Practical submission	A teacher saves a text response and an optional HTTPS resource link, then submits. Empty text is rejected; an invalid link does not silently disappear. Drafts remain editable.

Continuation of Table 12

ID / priority	Requirement	Acceptance criteria
FR07 / Must	Rubric feedback	An administrator reviews a submitted task using four 0-2 criteria, gives a comment and chooses accepted or revision needed. A teacher can see only their own feedback.
FR08 / Must	Progress	The dashboard distinguishes not started, in progress, awaiting review, revision needed and completed. A module completes only after the required lesson steps and an accepted practice submission.
FR09 / Must	Content administration	An administrator creates/edits drafts and publishes a new version. Existing attempts retain their original question and rubric version; publish does not rewrite history.
FR10 / Must	Data control	A teacher can obtain their own assessment/submission history and request account deletion. The administrator removes associated personal records under the agreed retention process; deletion is logged without retaining the content.
FR11 / Should	Resume and recover	A saved draft can be reopened after sign-out. A failed save shows an error, preserves visible text and offers retry without showing a false success.
FR12 / Should	Pilot export	An authorised administrator exports pseudonymised, consented pilot records with task/form version, scores, condition and timestamps. Non-consenting accounts are excluded.

6.1 Non-functional requirements

The quality requirements and their acceptance criteria are listed in Table 13.

Table 13 - Non-functional requirements and acceptance criteria

ID / priority	Attribute	Acceptance criteria
NFR01 / Must	Access control	Integration tests using two teachers and an admin show zero successful cross-user reads/writes. Authorisation is checked on the server for every protected resource.
NFR02 / Must	Accessibility	Core flows work by keyboard; labels and visible focus exist; errors are associated with fields. Test selected WCAG 2.2 A/AA criteria [12]; record failures rather than claiming full conformance.

Continuation of Table 13

ID / priority	Attribute	Acceptance criteria
NFR03 / Must	Responsive layout	Core screens at 360, 768 and 1280 CSS pixels have readable controls and no horizontal page scrolling; test keyboard and zoom behaviour.
NFR04 / Should	Performance target	On the recorded staging hardware, seed 100 teacher accounts and run 20 concurrent sessions for 5 minutes. Target p95 API response under 1 second and under 1% unexpected failures. Report actual results and exclusions.
NFR05 / Must	Data integrity	A failed transaction does not produce a completed attempt or duplicated progress. Concurrent/retried submission tests preserve one record per idempotency key.
NFR06 / Must	Reproducibility	A fresh checkout can be configured with example variables, migrated, seeded with synthetic data and tested using the documented commands; no private data or credentials in Git.
NFR07 / Must	Privacy	Collect only account and learning data needed for the workflow. Do not collect pupil records; keep contact keys separate from research exports. Agree retention before pilot consent.
NFR08 / Should	Recovery	Restore a staging backup into an isolated database and verify seeded counts and relationships. Target restoration within 60 minutes; record actual elapsed time.

6.2 Scoring and module selection

The initial diagnostic uses three original scenario questions for each module, scored 0 or 1. Module score = correct answers / 3. The recommended module is the incomplete module with the lowest score; equal scores follow the fixed order M1, M2, M3. A score below 2/3 is labelled "suggested starting point", not "low competence". Confidence prompts are not part of the numeric score. Teachers can override the order. If every module is completed, the page suggests review instead of inventing a new need. These thresholds are an initial product rule requiring review, not a validated psychometric boundary.

Completion has a separate meaning from improvement: completion records the required activities and an accepted practice submission, whereas improvement requires a comparison of practical performance over time. The application must not

infer improvement from time spent, a confidence response or a completed lesson alone.

6.3 Main use case and alternatives

UC2-UC5 cover the teacher learning cycle. Preconditions: the teacher is authenticated and three module versions are published. Main flow: answer the diagnostic; submit; read the recommendation; open a module; study; save and submit practice; receive feedback; inspect progress. Alternatives: skip personalisation and browse; resume a draft; correct an invalid link; retry a failed save; revise a returned submission. Postcondition: the saved status reflects the actual completed step, and the teacher can retrieve their own record.

7 Architecture and diagrams

The proposed architecture is a modular monolith: a Next.js application serves the interface and protected endpoints, while PostgreSQL stores accounts, versioned content and learning records. Business services sit behind the API boundary. The first version has no mandatory external learning-platform integration. An optional resource URL is stored as a link; the server does not fetch arbitrary user-supplied URLs. This avoids turning the submission feature into an unrestricted server-side URL fetcher.

Figure 1 shows the system boundary and its external actors.

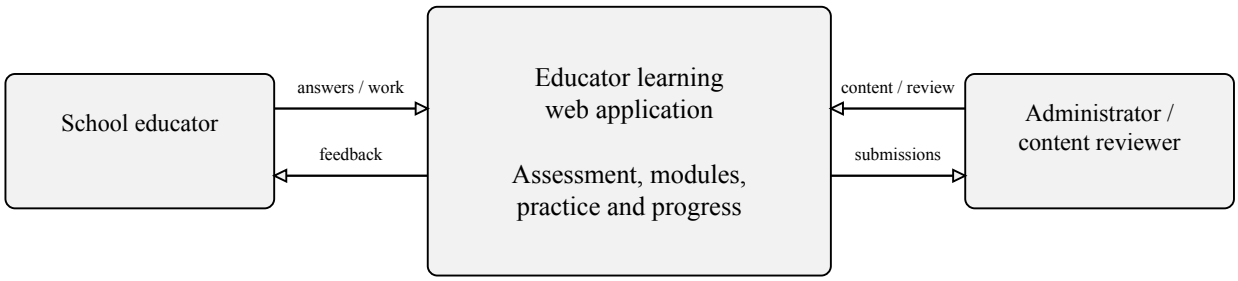


Figure 1 - System context. Arrows describe information exchanged with the application. Project-review access to Git and the tracker is outside the runtime system.

Figure 2 shows the interactions available to each application role.

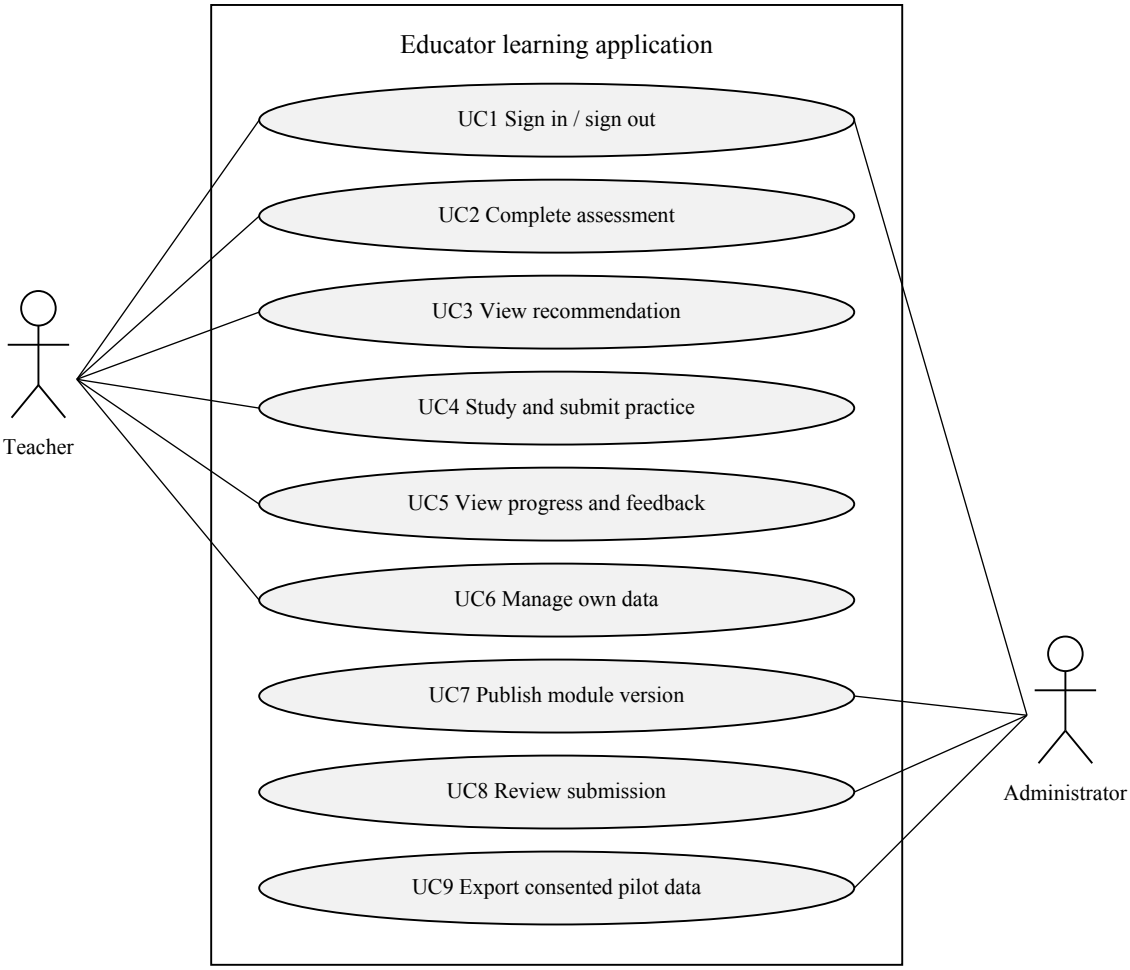


Figure 2 - Use case diagram. Teacher and administrator are application roles. Associations show permitted interactions; they do not imply that the supervisor operates the application.

Figure 3 shows the proposed server components and persistence boundary.

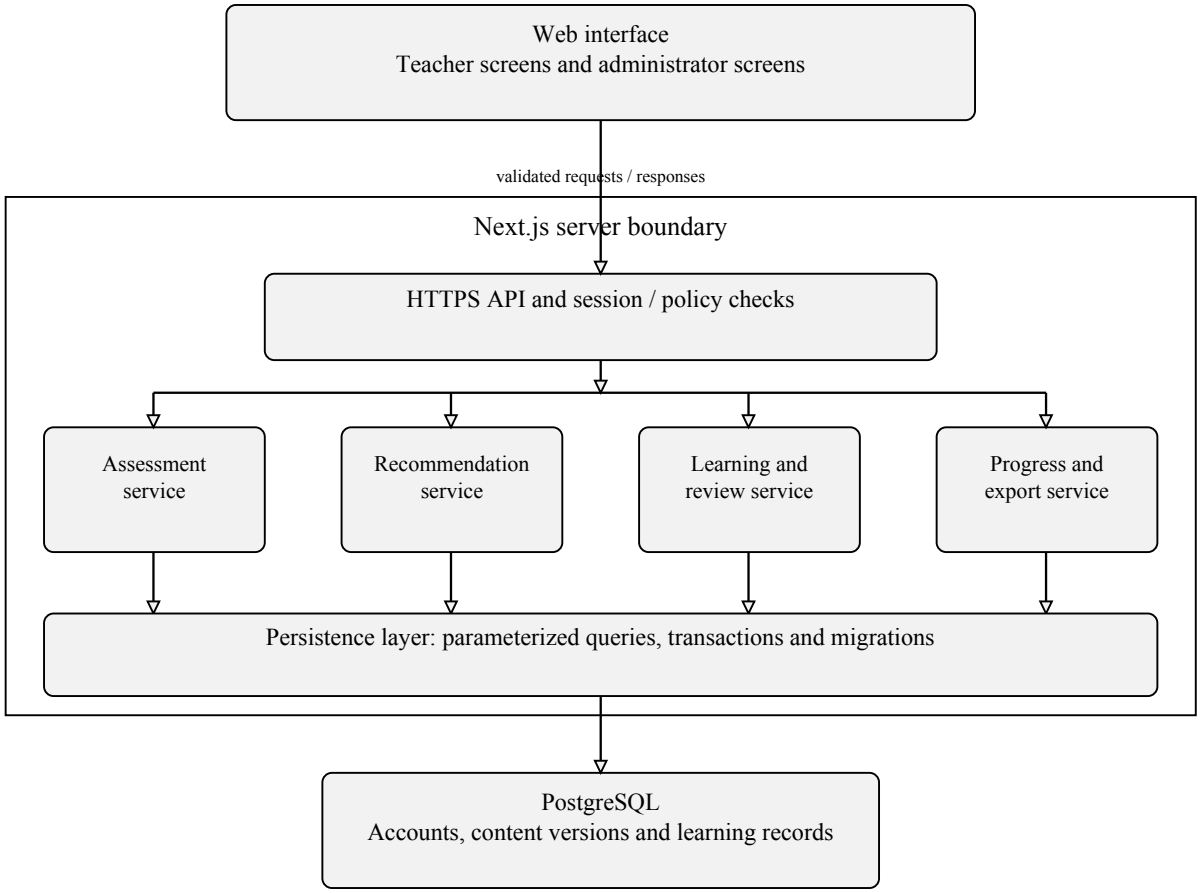


Figure 3 - Draft component diagram. Server-side services enforce authorisation and invariants before repository access. The database is not directly exposed to the browser.

7.1 Responsibilities and trust boundaries

Table 14 assigns responsibilities and requirements to the proposed components.

Table 14 - Component responsibilities and requirement traceability

Component	Responsibility	Requirements
Web interface	Assessment forms, module pages, practice and progress; show save/error states.	FR02, FR05-FR08, FR11; NFR02-NFR03
Auth and policy service	Establish session, role and resource ownership; deny access by default.	FR01, FR10; NFR01, NFR07
Assessment service	Validate answers, compute score and save a versioned attempt atomically.	FR02-FR03; NFR05
Recommendation service	Apply versioned deterministic rules; return module and explanation.	FR04

Continuation of Table 14

Component	Responsibility	Requirements
Learning and review service	Published content, drafts, submissions, rubric reviews and state transitions.	FR05-FR07, FR09, FR11
Progress and export service	Derive status from records; export own data or consented pilot data.	FR08, FR10, FR12
Persistence layer	Parameterized queries, transactions, foreign keys and migrations.	NFR05-NFR08

7.2 Initial data model

The initial entities and their invariants are summarised in Table 15.

Table 15 - Initial data model

Entity	Key data	Relationship or invariant
User	id, email, password hash, role	One account owns its attempts and submissions. Role is server-controlled.
ModuleVersion	id, module key, version, status, objective, content	Published versions are immutable; create another version to revise content.
Question	id, module version, prompt, options, correct option	Server retains answer key; client receives only public question fields.
AssessmentAttempt / Response	user id, version, score, question id, selected answer	Responses belong to one attempt; question versions remain recoverable.
Recommendation	attempt id, module key, reason, rule version	Recommendation is reproducible from its linked attempt and rules.
Submission	user id, module version, text, optional URL, status	Only the owner edits drafts; submitted revisions retain a version.
Review	submission id, reviewer id, four rubric scores, comment	Reviewer must be an administrator; score values are integers 0-2.
Progress / Consent	user id, module key, step; consent version/time	Progress is derived from durable activity records; pilot export requires consent.

7.3 Request and failure handling

A submitted assessment reaches an authenticated endpoint. The server validates question identifiers and answer counts, loads the matching version, computes the score, and writes the attempt and responses in a transaction. After commit, the recommendation service produces a reasoned next step. A network retry uses the same request key and returns the existing result. The client displays success only after server confirmation. The same ownership check applies when an existing record is requested by identifier.

Passwords will use a maintained authentication library and adaptive hashing; cookies will use secure session settings in deployment. Mutations need CSRF/origin protection appropriate to the selected session mechanism. Rich text must be sanitised or stored as plain text. Content links are validated as HTTPS and opened with safe browser attributes. Logging records identifiers and error categories rather than lesson responses or account credentials. These are implementation obligations, not claims that security tests have passed.

8 UX prototype

The supplied prototype is a local HTML interface with simulated data and no backend. It demonstrates the navigation and decision states, not a deployed application. It contains the teacher dashboard, a nine-question diagnostic interaction, recommendation, learning module, practice submission and administrator review. Demonstration scores and feedback are explicitly labelled; they are not participant results. The diagnostic covers three questions per topic, with confidence responses recorded separately, as specified in FR02.

Table 16 identifies the main screens and the states each screen must communicate.

Table 16 - Key user screens and states

Screen	Purpose	Requirement and important state
Dashboard	Explain the next step and show separate activity statuses.	FR08; empty assessment and completed module states.
Assessment	Select an answer and understand what is being measured.	FR02-FR03; unanswered question error; progress indication.
Recommendation	Show the suggested topic and a readable reason; permit another choice.	FR04; no misleading certification label.
Module	Read an objective, worked example and practice brief.	FR05; return to catalogue without losing state.
Practice	Draft a response, optionally add a link and submit.	FR06, FR11; empty text and invalid URL errors.
Progress and review	Distinguish awaiting review from completed; inspect rubric feedback.	FR07-FR08; administrator role switch is demonstration only.

Figure 4 shows the proposed teacher dashboard.

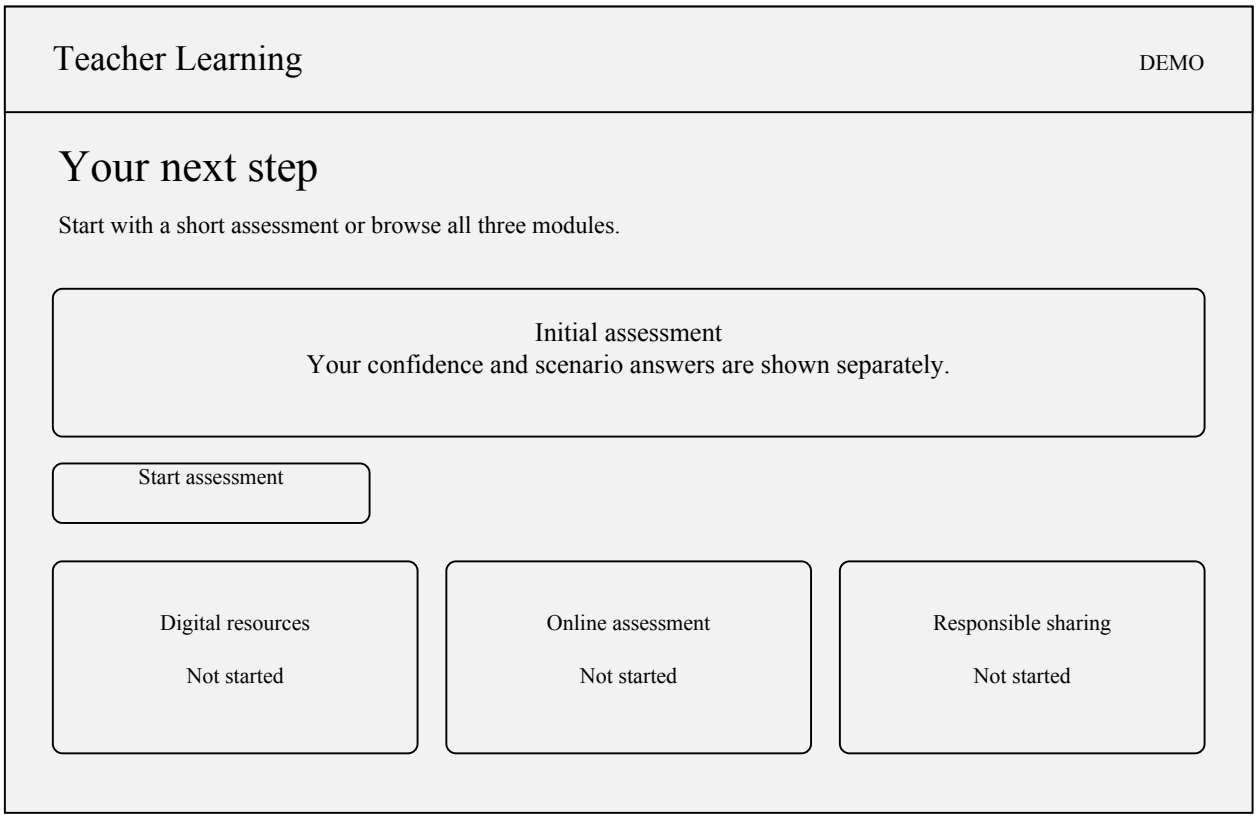


Figure 4 - Dashboard design. Demonstration content only; the interactive source is in repository/prototype/index.html.

Figure 5 shows the proposed practical-work submission screen.

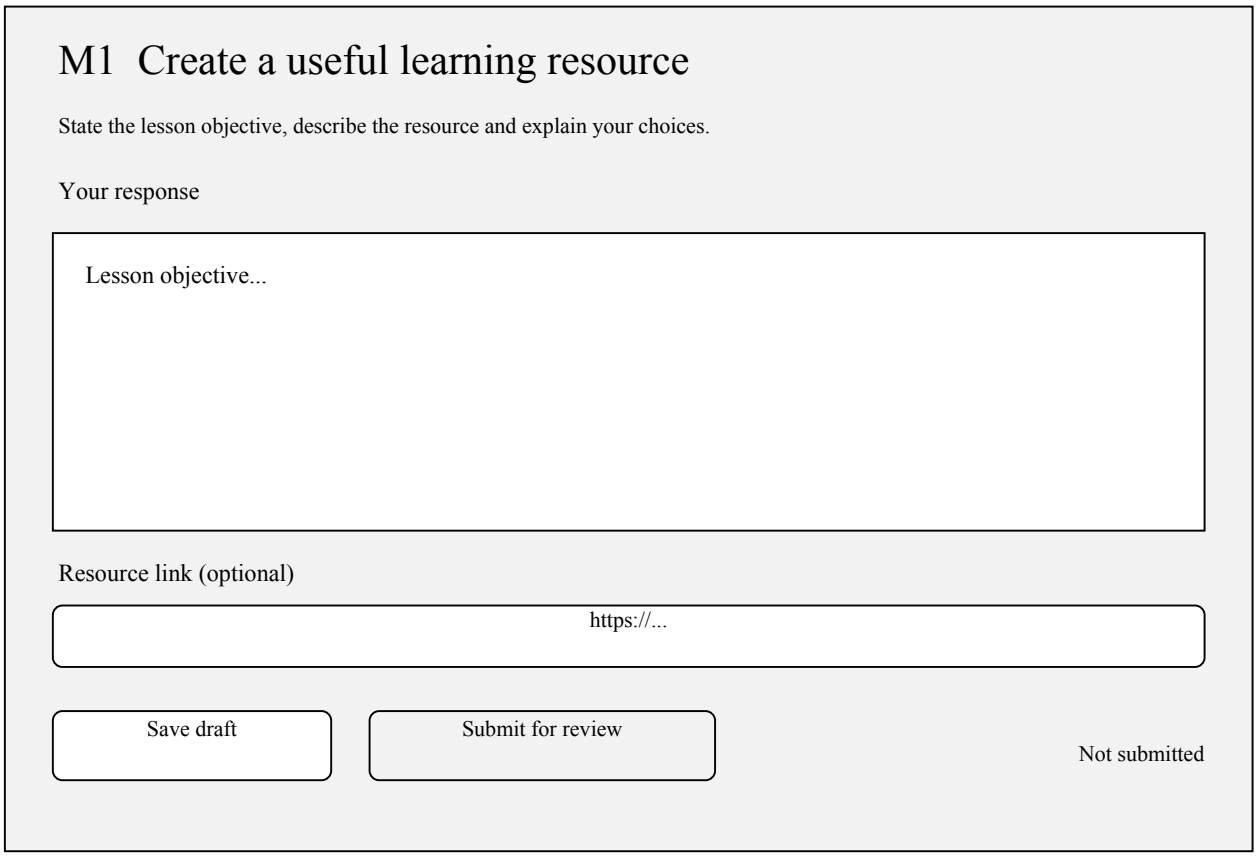


Figure 5 - Practice submission design. The response is saved as a draft before review; completion is not awarded merely for opening a lesson.

8.1 Interaction decisions to validate

Use one clear primary action per screen and a persistent route back to the module catalogue. Present the task objective before the tool instructions so that the educator knows what the activity is for. Keep confidence, diagnostic correctness and practice feedback visibly separate. Avoid public rankings of teachers. A score explanation should be a sentence that names the underlying answers, not an unexplained coloured badge.

The report and demonstration interface use English for the university review. Russian and Kazakh content requirements are still unconfirmed. Interface strings should remain separate from content so localisation can follow user discussions. A successful walkthrough by the student is only a technical check of the prototype. Usability with teachers must be observed and recorded separately.

The browser checks in Appendix B use synthetic examples. They verify the interaction design and do not establish server security, persistent storage or learning effectiveness.

9 Verification and evaluation plan

9.1 Questions and baseline

Verification asks whether the application fulfils its requirements; evaluation asks whether the intended workflow is usable and whether participants show preliminary changes in practical performance. The pilot baseline is a static page with the same module content and task briefs, without diagnosis-based ordering or a progress dashboard. Content and task difficulty should be matched as closely as possible. Both conditions must be documented so differences are not attributed only to the interface when content also differs.

Recruitment target for a later formative pilot is 5-8 adult school educators, subject to availability and the supervisor's approval of the protocol. This is a proposed target, not a recruited sample or a statistically powered efficacy study. The 3-5 initial requirements discussions serve a different purpose and should not be relabelled as a learning trial. Record any overlap between participants.

The planned verification scenarios and expected results are specified in Table 17.

Table 17 - Verification scenarios

Test	Trace	Method	Expected result
TC01	FR01, NFR01	Use unauthenticated, teacher A, teacher B and admin requests. Attempt cross-user access and admin mutation.	401/403 as appropriate; no protected data or changes.
TC02	FR02-FR03	Submit incomplete, valid and tampered answer sets; retry a valid request twice.	Missing/tampered input rejected; valid score correct; one saved attempt.
TC03	FR04	Test scores 0/3, 1/3, 2/3, 3/3, ties, completed modules and absent attempt.	Documented ordering and reason; no arbitrary recommendation.
TC04	FR05, FR09	Publish v2 after a teacher completed an attempt on v1.	Teacher history still displays v1; new sessions receive published v2.
TC05	FR06, FR11	Save a draft; reload; submit empty text or a malformed link; simulate network failure.	Draft recoverable; errors visible; failure never shown as successful submission.

Continuation of Table 17

Test	Trace	Method	Expected result
TC06	FR07-FR08	Submit practice, request revision, resubmit and accept after all required steps.	Awaiting review and revision states preserved; completion only when conditions are met.
TC07	FR10, FR12	Export own data and consented pilot records; test another user's identifier.	Only authorised/consented records returned; disallowed access denied.
TC08	NFR02-NFR 03	Keyboard-only walkthrough, screen-reader labels, 200% zoom and 360/768/1280 widths.	No unreachable control or hidden focus; errors announced; no page overflow.
TC09	NFR04	Run defined 20-session staging workload after warm-up; retain logs and dataset size.	Report p50/p95 latency and errors against target; do not omit failed requests.
TC10	NFR05, NFR08	Inject a transaction failure and restore a backup into an isolated database.	No partial completed record; restored references/counts match expected data.
TC11	NFR06	Fresh checkout on a clean environment using documented configuration.	Migration, seed and selected tests succeed without private files.

9.2 Measurement conditions

For a formative usability comparison, each teacher completes two comparable navigation tasks: find a relevant module and locate the next practice step. Alternate which interface is shown first and assign different but matched topics to reduce direct repetition. Record device, browser, connection problems, previous platform familiarity, task version, start/end time and every intervention by the moderator. Stop a task after 10 minutes and record a non-completion instead of discarding it. This small comparison cannot eliminate order or novelty effects.

Table 18 distinguishes usability measures from learning-related observations.

Table 18 - Evaluation measures and interpretation

Measure	Definition and collection	Interpretation
Task completion	Completed without moderator help / all attempted tasks; also show raw counts.	Proposed formative target: at least 80%. Report assisted completion separately.
Time on task	Seconds from the task prompt to correct completion; include individual times and non-completions.	Use median and range for completed tasks; do not treat failure as a fast time.
Observed difficulty	Count and describe wrong turns, requests for help and blocked steps.	Prioritise recurring problems and severe single failures.
Perceived ease	After each task, one 1-7 ease question plus an optional explanation.	Project-specific rating, not a validated overall usability scale.
Practical performance	Four rubric criteria, each 0-2, on parallel pre/post tasks; maximum 8.	Report each paired change and median change. A target of +1 point is provisional, not proof of causality.
Retention signal	If feasible, a parallel task one week later; record attrition.	Exploratory only; do not claim durable learning if follow-up is missing.

9.3 Practical rubric and bias control

The proposed four-criterion rubric is shown in Table 19.

Table 19 - Practical task scoring rubric

Criterion	0 points	1 point	2 points
Pedagogical fit	No clear lesson objective or unsuitable activity.	Objective stated, but link to activity is incomplete.	Activity clearly serves the objective and intended learners.
Technical correctness	Essential function missing or unusable.	Main function works with a minor gap.	Specified resource/quiz/sharing action is complete and usable.
Responsible and accessible use	Major privacy, attribution or access issue.	Relevant issue identified but incompletely addressed.	Appropriate attribution, access and privacy choices explained.

Continuation of Table 19

Criterion	0 points	1 point	2 points
Reasoning and feedback	No explanation of choices or use of results.	Partial explanation without a clear next action.	Choices justified and feedback/next action follows from evidence.

Use two parallel task forms rather than exactly the same resource before and after a lesson. Freeze the rubric before collecting outcomes. Where possible, a second reviewer scores anonymised artefacts without seeing whether they are pre- or post-module. Record disagreements and resolution; do not claim independent review if only the student scores the work. Report all consenting participants, missing pairs and withdrawals. With a small convenience sample and no randomised learning control, score changes are preliminary observations, not a causal estimate of the application's effectiveness.

9.4 Release gates

Before a pilot: resolve all critical access-control and data-loss defects; pass mandatory functional scenarios; obtain informed consent; check the three module tasks and rubric with the supervisor; test account isolation and export scope. Before final evaluation: freeze the application version and content versions and document the baseline. No application or learning tests have been executed at this first-report stage; checks of the supplied local prototype are reported separately in prototype/verification.txt.

10 Project management and risks

The project is individual. Ayan owns all work items; supervisor review is an external dependency, not an assigned implementation task. The final Justification places formal supervision and a collaborative schedule in the third trimester of 2026-2027. Earlier draft justifications proposed a different cadence; this report follows the final version. The student confirmed on 23 October that the university has not provided the next pre-defence or final-defence dates. Only first-submission preparation targets are dated; the later milestones below describe dependency order, not an agreed calendar.

Table 20 separates dated first-submission targets from later milestones without an agreed calendar.

Table 20 - Milestones and exit conditions

Milestone	Target	Exit condition
First-submission documents	22-23 October 2026	Reviewable report, diagrams, prototype and task breakdown prepared.
Primary evidence and review	Before final packaging on 24 October	Actual discussions or client agreement; substantive comments and recorded responses. High risk if participants/reviewer unavailable.
Access and release verification	24 October, before 17:00 Astana	Remote repository/tracker verified, authentic screenshots, all recipients can open materials.
Submit with buffer	24 October, by 17:30 Astana	Required form fields completed and receipt saved; official deadline remains 18:00.
Implementation planning	Date pending	Agree calendar, validated scope and review checkpoints with supervisor.
Core vertical workflow	After scope and calendar agreement	Accounts, one assessment-to-practice path, content versions and integration tests.
Remaining modules and reliability	After core workflow	Three complete modules, reviewer workflow, accessibility fixes and recovery checks.
Pilot and thesis revision	After release gates and consent	Consent, actual observations, analysis with limitations, corrections and final report.

10.1 Work breakdown and task traceability

The project contains 40 actual YouTrack issues, EDU-1 to EDU-40. The first 25 were created on 22 October; 15 development and evaluation tasks were added on 23 October. The Diploma tag groups 30 research, design, implementation, testing and evaluation tasks. The Submission tag groups 10 preparation and hand-in tasks. The main board filters Diploma; a separate saved search shows Submission. Local references W01-W40 map to actual issue IDs in project-management/issue-map.csv; the two numbering sequences are not always identical.

Every issue has an owner, priority, expected outcome, acceptance criteria and dependency references. All work belongs to Ayan's admin account. The 19 later implementation, evaluation and final-thesis tasks remain unscheduled because university dates are unavailable. This is an explicit planning gap against the deadline requirement; dates must be agreed later. New application tasks remain in the backlog. Prepared documents remain in review; this does not establish supervisor approval.

The repository should have one permanent remote, a reviewed main branch and task branches. Include the real tracker ID in each branch and commit or pull-request description, and link the change back from the task. Significant design and evaluation changes need selected supervisor reviews at agreed checkpoints. Do not create fake historical commits or backdate work. A repository created on 22 October can honestly show work from that date; it cannot demonstrate earlier progressive development.

10.2 Risks and responses

The principal risks, consequences and responses are recorded in Table 21.

Table 21 - Project risks and responses

Risk	Impact / likelihood	Response and trigger
No discussion participants before deadline	High / high	Recruit immediately; keep missing evidence explicit. A written protocol alone does not satisfy the discussions requirement.

Continuation of Table 21

Risk	Impact / likelihood	Response and trigger
No supervisor feedback in time	High / high	Prepare a focused review pack. Do not label the new report approved without comments or confirmation.
Repository/tracker or committee access missing	High / high	Create and verify project links early; obtain exact recipient accounts. Invite acceptance may still be pending.
Scope expands beyond three modules	High / medium	Defer optional features; preserve the full assessment-practice-progress workflow.
Self-report mistaken for learning	High / medium	Keep confidence, quiz scores and rubric outcomes separate; limit claims.
Participant data enters Git	High / medium	Use coded notes, synthetic examples and separate private storage; inspect changes before push.
Task review is inconsistent	Medium / medium	Freeze and calibrate rubric; record scoring disagreement and review capacity.
Late form or upload failure	High / medium	Prepare two PDFs and links in advance; use a 30-minute buffer and retain receipt.

10.3 Repository and tracker acceptance audit

The minimum repository content is provided locally: README, report sections, comparisons, requirements, editable diagrams, prototype and evaluation plan. The private GitHub repository is https://github.com/ayanvrsn/educator-digital-competencies and the YouTrack project is https://ayan-educator-diploma.youtrack.cloud/projects/EDU. Both were created on 22 October 2026. A local Git author identity and the main remote have been configured; the review materials are developed on the task branch EDU-13/first-submission-preparation. Supervisor review and committee access remain pending. The live form requires prob-period-1, while the requirements DOCX names pre-defense-1. Create both tags on the same exact submission commit after final checks; use the prob-period-1 link in the form and do not move either submitted tag. The CSV files are portable copies of the plan; the live YouTrack project is the task record. Recipient access still needs an independent check.

Issues were created directly in the YouTrack interface. The supplied CSV is a backup for portability, not evidence of a CSV import. If a future import is needed, the documented CSV workflow uses a Google spreadsheet intermediary [26]. The accompanying YouTrack_Task_Screenshots.pdf contains authentic captures from 23 October 2026: the 30-task Diploma board, the separate 10-task Submission view, implementation issues EDU-31 and EDU-30. The cards show ownership, priority, acceptance criteria and either a preparation target or an explicitly unset future date. These captures record preparation status, not supervisor approval or completed interviews; refresh them before final submission.

The repository setup follows the GitHub documentation [27]. The current review package is available in draft pull request 1. Appendix A reproduces selected tracker views; Appendix B records the checks actually performed on the local prototype; Appendix C lists the 40 work items and their current planning states.

11 Supervisor feedback

No substantive feedback on the deliverables in this report is available. The student confirmed this on 22 October. The email confirming the Justification and its dispatch establishes supervision context, but does not establish that the supervisor reviewed this literature synthesis, architecture, prototype or evaluation protocol. No approval, comment or signature has been invented.

Table 22 is an agenda for obtaining feedback; it does not represent feedback already received.

Table 22 - Supervisor review agenda

Review request	Material to inspect	Decision needed
Scope and competence mapping	Sections 1, 3 and module map	Are the three modules and selected competences suitable for the diploma?
Evidence and requirements	Sections 2 and 6; actual discussion notes when available	Do the proposed requirements follow from sufficient evidence?
Architecture and UX	Sections 7-8 and local prototype	Are the workflow and component boundaries feasible for one student?
Evaluation	Section 9; baseline and rubric	Are the pilot, consent, parallel tasks and limits of claims appropriate?

A feedback log template is supplied separately with fields for the date, exact comment or faithful paraphrase, source, decision, changed artefact and verification. Its current state is "awaiting supervisor feedback". After review, record rejected or deferred suggestions with reasons as well as accepted changes. Do not mark a review complete merely because a request was sent.

CONCLUSION

The first stage has produced a defined project scope, a literature synthesis, comparisons, an initial specification, three architecture diagrams, a reviewable interface prototype and an evaluation plan. The design connects diagnostic questions to three learning modules and practical work, with confidence, quiz results and rubric scores kept separate.

These results support the next design review and implementation planning. They do not yet demonstrate that the application improves educators' digital competence. Local needs remain unverified without the required educator discussions, and the substantive supervisor review is still missing.

The next actions are to collect the primary evidence, revise the requirements where necessary, obtain and record supervisor comments, and verify recipient access. Implementation can then proceed through the dependencies recorded in YouTrack. Pilot data must be collected only after the release gates and consent requirements are met. No economic benefit or learning effect is claimed without measurement.

REFERENCES

The core review contains 12 sources, including eight scientific publications: [6], [8], [3], [2], [7], [4], [9], [10]. The remaining references support product and engineering decisions. Online source checks were performed on 22 October 2026.

1 Redecker C. European Framework for the Digital Competence of Educators: DigCompEdu / ed. Y. Punie. - Luxembourg: Publications Office of the European Union, 2017. - DOI: 10.2760/159770. - URL: https://publications.jrc.ec.europa.eu/repository/handle/JRC107466 (accessed: 22.10.2026).

2 Teachers' digital competencies in higher education: a systematic literature review / V. Basilotta-Gómez-Pablos, M. Matarranz, L.-A. Casado-Aranda, A. Otto // International Journal of Educational Technology in Higher Education. - 2022. - Vol. 19. - Art. 8. - URL: https://doi.org/10.1186/s41239-021-00312-8 (accessed: 22.10.2026).

3 Design and Validation of t-MOOC for the Development of the Digital Competence of Non-University Teachers / J. Cabero-Almenara, R. Barragán-Sánchez, A. Palacios-Rodríguez, L. Martín-Párraga // Technologies. - 2021. - Vol. 9, no. 4. - Art. 84. - URL: https://doi.org/10.3390/technologies9040084 (accessed: 22.10.2026).

4 Systematic review of quantitative research on digital competences of in-service school teachers / M. Claro, C. Castro-Grau, J. M. Ochoa et al. // Computers & Education. - 2024. - Vol. 215. - Art. 105030. - URL: https://doi.org/10.1016/j.compedu.2024.105030 (accessed: 22.10.2026).

5 UNESCO ICT Competency Framework for Teachers. Version 3 / UNESCO. - Paris: UNESCO, 2018. - ISBN 978-92-3-100285-4. - URL: https://unesdoc.unesco.org/ark:/48223/pf0000265721 (accessed: 22.10.2026).

6 Falloon G. From digital literacy to digital competence: the teacher digital competency (TDC) framework // Educational Technology Research and Development. - 2020. - Vol. 68. - P. 2449-2472. - URL: https://doi.org/10.1007/s11423-020-09767-4 (accessed: 22.10.2026).

7 Heine S., Krepf M., König J. Digital resources as an aspect of teacher professional digital competence: One term, different definitions - a systematic review // Education and Information Technologies. - 2023. - Vol. 28. - P. 3711-3738. - First published online in 2022. - URL: https://doi.org/10.1007/s10639-022-11321-z (accessed: 22.10.2026).

8 The relation between in-service teachers' digital competence and personal and contextual factors: What matters most? / M. Lucas, P. Bem-Haja, F. Siddiq et al. // Computers & Education. - 2021. - Vol. 160. - Art. 104052. - URL: https://doi.org/10.1016/j.compedu.2020.104052 (accessed: 22.10.2026).

9 Tzafilkou K., Perifanou M., Economides A. A. Assessing teachers' digital competence in primary and secondary education: Applying a new instrument to integrate pedagogical and professional elements for digital education // Education and Information Technologies. - 2023. - Vol. 28. - P. 16017-16040. - URL: https://doi.org/10.1007/s10639-023-11848-9 (accessed: 22.10.2026).

10 Gallego Joya L., Merchán Merchán M. A., López Barrera E. A. Development and strengthening of teachers' digital competence: Systematic review // Contemporary Educational Technology. - 2025. - Vol. 17, no. 1. - Art. ep555. - URL: https://doi.org/10.30935/cedtech/15744 (accessed: 22.10.2026).

11 Cosgrove J., Cachia R. DigComp 3.0: European Digital Competence Framework. - 5th ed. - Luxembourg: Publications Office of the European Union, 2025. - DOI: 10.2760/0001149. - URL: https://publications.jrc.ec.europa.eu/repository/handle/JRC144121 (accessed: 22.10.2026).

12 Web Content Accessibility Guidelines (WCAG) 2.2 [Electronic resource] / W3C. - W3C Recommendation, 12 December 2024. - URL: https://www.w3.org/TR/2024/REC-WCAG22-20241212/ (accessed: 22.10.2026).

13 European Commission. How SELFIE for TEACHERS works. Official product documentation [Electronic resource]. - URL: https://education.ec.europa.eu/selfie-for-teachers/get-started (accessed: 22.10.2026).

14 Google for Education. Training and Professional Development [Electronic resource]. - URL:

https://edu.google.com/resources/get-started/professional-development/ (accessed: 22.10.2026).

15 Google for Education. Certifications for Educators [Electronic resource]. - URL: https://edu.google.com/intl/ALL_us/learning-center/certifications/ (accessed: 22.10.2026).

16 Microsoft Learn. Educator Center [Electronic resource]. - URL: https://learn.microsoft.com/en-us/training/educator-center/ (accessed: 22.10.2026).

17 Microsoft Learn. Frequently asked questions [Electronic resource]. - URL: https://learn.microsoft.com/en-us/training/support/faq (accessed: 22.10.2026).

18 Next.js. Route Handlers [Electronic resource]. - URL: https://nextjs.org/docs/app/getting-started/route-handlers (accessed: 22.10.2026).

19 Django. Django at a glance. Documentation 5.2 [Electronic resource]. - URL: https://docs.djangoproject.com/en/5.2/intro/overview/ (accessed: 22.10.2026).

20 NestJS. Documentation [Electronic resource]. - URL: https://docs.nestjs.com/ (accessed: 22.10.2026).

21 PostgreSQL. Transactions. Documentation 18 [Electronic resource]. - URL: https://www.postgresql.org/docs/18/tutorial-transactions.html (accessed: 22.10.2026).

22 SQLite. Appropriate Uses for SQLite [Electronic resource]. - URL: https://www.sqlite.org/whentouse.html (accessed: 22.10.2026).

23 MongoDB. Data Modeling [Electronic resource]. - URL: https://www.mongodb.com/docs/manual/data-modeling/ (accessed: 22.10.2026).

24 GraphQL. Learn GraphQL [Electronic resource]. - URL: https://graphql.org/learn/ (accessed: 22.10.2026).

25 Next.js. use server directive [Electronic resource]. - URL: https://nextjs.org/docs/app/api-reference/directives/use-server (accessed: 22.10.2026).

26 JetBrains. Import from CSV. YouTrack Cloud 2026.2 Help [Electronic resource]. - URL: https://www.jetbrains.com/help/youtrack/cloud/import-from-csv.html (accessed: 22.10.2026).

27 GitHub Docs. Creating a new repository [Electronic resource]. - URL: https://docs.github.com/en/repositories/creating-and-managing-repositories/creating-a-new-repository (accessed: 22.10.2026).

APPENDIX A

(informative)
Tracker evidence

Figures A.1-A.4 reproduce authentic browser captures from 23 October 2026. They show the separate Diploma and Submission views and two implementation tasks. The screenshots are included on A3 sheets to preserve readability. The same evidence is also supplied in the separate tracker PDF required for submission.

The main board contains 30 Diploma tasks: 21 in the backlog and 9 in review. The Submission view contains 10 organisational tasks. Review is a workflow state and does not establish supervisor approval. Nineteen future tasks have no assigned calendar date because the university has not provided the next pre-defence or defence schedule.

Project: https://ayan-educator-diploma.youtrack.cloud/projects/EDU

Repository: https://github.com/ayanvrsn/educator-digital-competencies

Draft review: https://github.com/ayanvrsn/educator-digital-competencies/pull/1

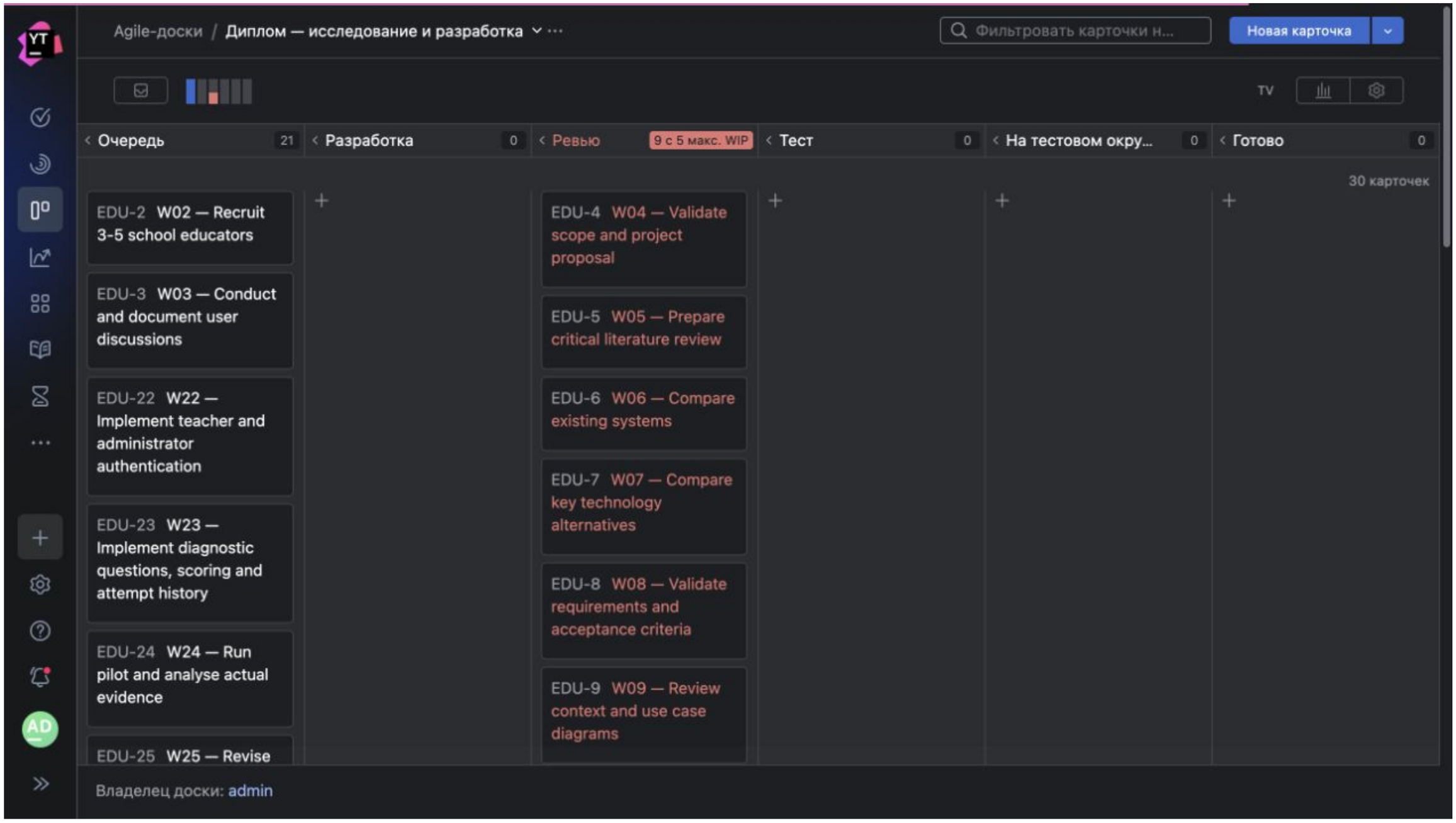


Figure A.1 - The Diploma board: research, implementation and evaluation

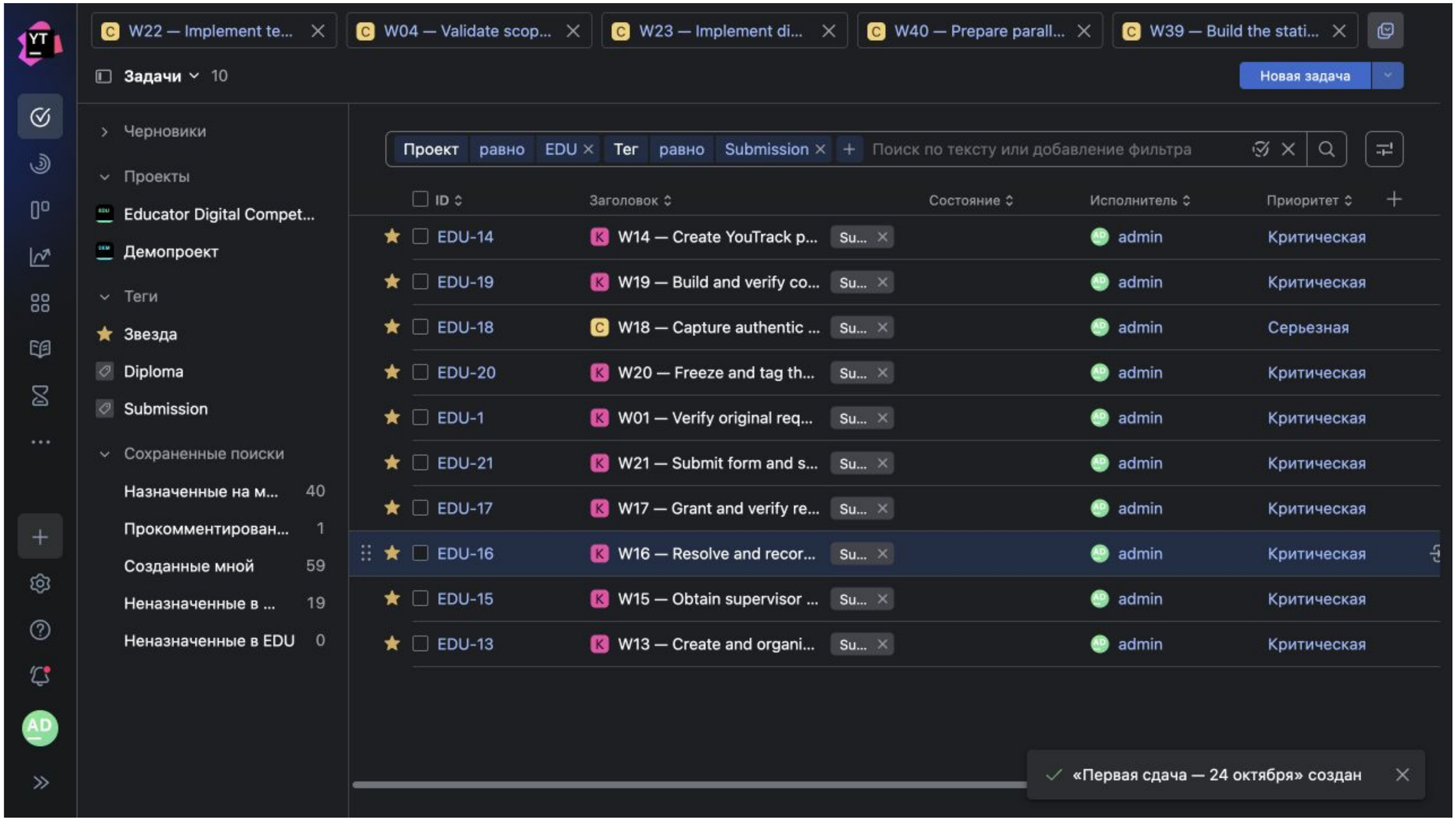


Figure A.2 - The separate Submission task view

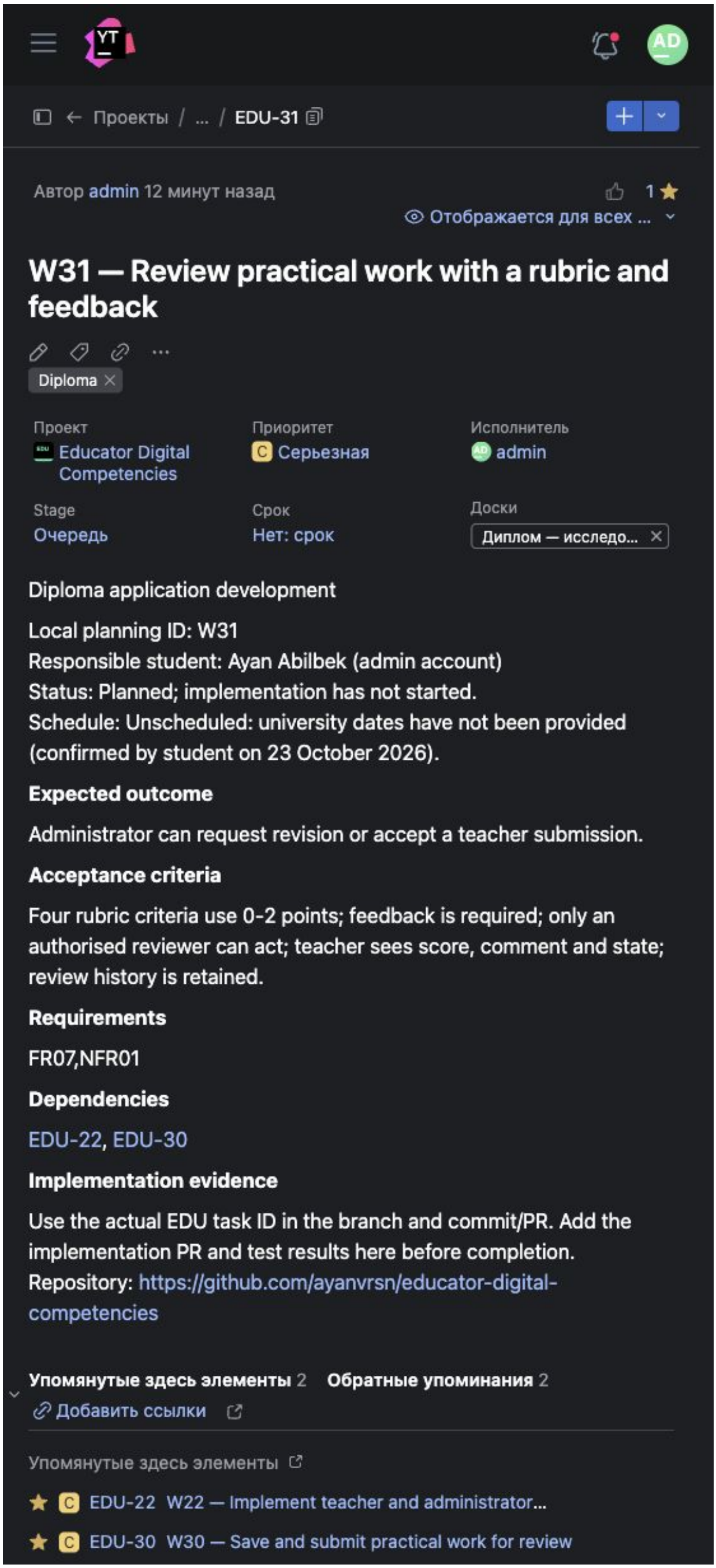


Figure A.3 - EDU-31: rubric-based review and feedback implementation

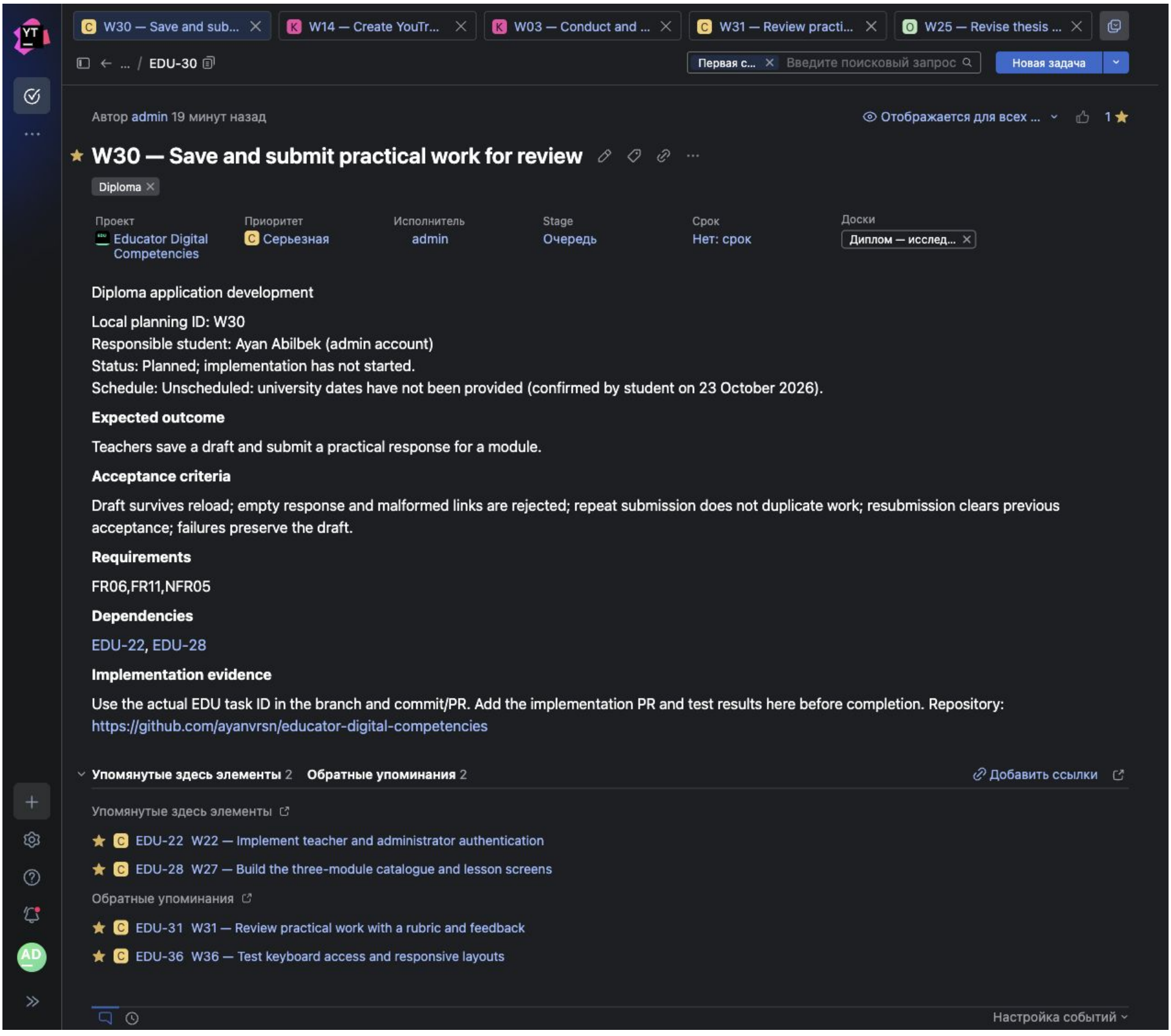


Figure A.4 - EDU-30: save and submit practical work

APPENDIX B

(informative)
Observed prototype checks

Scope: manual browser walkthrough in the Codex in-app browser using synthetic examples only. These checks concern the supplied HTML prototype, not an implemented backend or a teacher usability study.

The following behaviour was observed using synthetic examples:

- An unanswered diagnostic item prevents advancing and displays a clear message.
- Nine scenario questions lead to three separate confidence responses.
- A synthetic answer pattern M1=0/3, M2=3/3, M3=3/3 recommends M1. Confidence remains separate.
- Marking the lesson step complete changes progress but does not complete the module.
- Empty practice submission is rejected.
- An HTTP resource link is rejected; leaving the optional link empty is accepted.
- Saving a draft preserves the synthetic response when navigating to Progress and back within the same page session.
- Submitting changes the state to Awaiting review.
- The administrator demo requires explanatory feedback before accepting a submission.
- Four synthetic rubric ratings 2+2+2+1 display 7/8; acceptance plus the completed lesson produces Completed.
- Re-submitting accepted practice returns to Awaiting review. Clicking the lesson completion control does not restore the old acceptance.

Defect fixed during preparation: stale acceptance could complete a re-submitted task. Acceptance is now cleared on draft save and resubmission.

Compact layout: a browser screenshot at an observed CSS width of 599 px showed no horizontal overflow (scrollWidth = clientWidth = 599). A requested 360 px override did not take effect, so 360 px behaviour is not claimed as tested.

Limitations: no server authentication, database, role-isolation, performance or real learning evaluation was tested. No participant results are present. Reload intentionally clears the demonstration session. Screenshots of progress contain clearly marked synthetic data.

APPENDIX C

(informative)
Project work breakdown

Table C.1 lists the 40 actual YouTrack issues. Ayan Abilbek owns all tasks. Diploma covers research, design, implementation and evaluation; Submission covers the first hand-in. The dates are preparation targets, except that future implementation and evaluation dates remain unset. Full outcomes and acceptance criteria are recorded in the live issues and project-management/tasks.csv. Dependencies are issue mentions in descriptions.

Priority abbreviations: C - Critical; H - High; N - Normal. Review means awaiting checks. Blocked is the planning status; these tasks remain in the board backlog.

Table C.1 - Project work items

Issue	Work item / group	Priority / status	Target date
EDU-1	Verify original requirements and final justification Submission	C / Review	2026-10-22
EDU-2	Recruit 3-5 school educators Diploma	C / Open	2026-10-23
EDU-3	Conduct and document user discussions Diploma	C / Blocked	2026-10-23
EDU-4	Validate scope and project proposal Diploma	H / Review	2026-10-23
EDU-5	Prepare critical literature review Diploma	H / Review	2026-10-23
EDU-6	Compare existing systems Diploma	H / Review	2026-10-23
EDU-7	Compare key technology alternatives Diploma	H / Review	2026-10-23
EDU-8	Validate requirements and acceptance criteria Diploma	H / Review	2026-10-23
EDU-9	Review context and use case diagrams Diploma	H / Review	2026-10-23

Continuation of Table C.1

Issue	Work item / group	Priority / status	Target date
EDU-10	Review component architecture and data model Diploma	H / Review	2026-10-23
EDU-11	Build and check UX prototype Diploma	H / Review	2026-10-23
EDU-12	Prepare verification and evaluation plan Diploma	H / Review	2026-10-23
EDU-13	Create and organise permanent Git repository Submission	C / In Progress	2026-10-23
EDU-14	Create YouTrack project and task board Submission	C / Review	2026-10-23
EDU-15	Obtain supervisor feedback Submission	C / Blocked	2026-10-24
EDU-16	Resolve and record supervisor comments Submission	C / Blocked	2026-10-24
EDU-17	Grant and verify recipient access Submission	C / Blocked	2026-10-24
EDU-18	Capture authentic tracker screenshots Submission	H / Review	2026-10-24
EDU-19	Build and verify consolidated report Submission	C / Review	2026-10-24
EDU-20	Freeze and tag the first submission Submission	C / Blocked	2026-10-24
EDU-21	Submit form and save receipt Submission	C / Blocked	2026-10-24
EDU-22	Implement teacher and administrator authentication Diploma	H / Open	Not scheduled
EDU-23	Implement diagnostic questions, scoring and attempt history Diploma	H / Open	Not scheduled
EDU-24	Run pilot and analyse actual evidence Diploma	H / Open	Not scheduled

Continuation of Table C.1

Issue	Work item / group	Priority / status	Target date
EDU-25	Revise thesis and prepare defence Diploma	N / Open	Not scheduled
EDU-26	Create PostgreSQL schema and migration workflow Diploma	H / Open	Not scheduled
EDU-28	Build the three-module catalogue and lesson screens Diploma	H / Open	Not scheduled
EDU-27	Implement versioned content publishing for administrators Diploma	H / Open	Not scheduled
EDU-29	Recommend a starting module from diagnostic results Diploma	H / Open	Not scheduled
EDU-30	Save and submit practical work for review Diploma	H / Open	Not scheduled
EDU-31	Review practical work with a rubric and feedback Diploma	H / Open	Not scheduled
EDU-32	Show persistent learning progress and review states Diploma	H / Open	Not scheduled
EDU-33	Implement teacher data export and deletion request Diploma	H / Open	Not scheduled
EDU-34	Export anonymised pilot data with consent controls Diploma	H / Open	Not scheduled
EDU-35	Test authorization, ownership and submission consistency Diploma	H / Open	Not scheduled
EDU-36	Test keyboard access and responsive layouts Diploma	H / Open	Not scheduled

Continuation of Table C.1

Issue	Work item / group	Priority / status	Target date
EDU-38	Measure performance and verify backup recovery Diploma	H / Open	Not scheduled
EDU-37	Prepare reproducible application setup and staging deployment Diploma	H / Open	Not scheduled
EDU-39	Build the static resource baseline for evaluation Diploma	H / Open	Not scheduled
EDU-40	Prepare parallel learning tasks and calibrate the rubric Diploma	H / Open	Not scheduled